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ORIGINAL PAPER

Professional Development Improves Teachers' Embedded Instruction and Children's Outcomes in a Chinese Inclusive Preschool

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Abstract

This study examined the effects of a professional development (PD) intervention on inservice teachers' use of embedded instruction practices within inclusive preschool classrooms. The results of a multiple baseline across participants design suggested the PD intervention was effective in increasing teachers' implementation fidelity of embedded instruction learning trials (EILTs) to address individualized learning objectives of preschoolers with autism spectrum disorder. Teachers generalized and maintained their use of EILTs to novel children with disabilities within novel settings during follow-up sessions three months after the PD ended. Positive outcomes of individualized learning objectives were also observed across children.

Keywords Inservice teacher preparation · Coaching · Embedded instruction · Inclusive education · Autism spectrum disorders

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Children with disabilities participating in inclusive education must receive specially designed instruction or intervention to address their unique developmental and learning needs (Division for Early Childhood & National Association for the Education of Young Children, 2009). Participation alone is not sufficient for children with disabilities to make meaningful progress in inclusive settings (Odom et al., 2002). Researchers and inclusive education advocates have stressed the importance of applying evidence-based practices (EBPs) to improve the quality of inclusive education (Barton & Smith, 2015). One EBP—embedded instruction (EI) can be used by classroom teachers to provide learning opportunities for an individual child during typical classroom routines and activities (Snyder et al., 2015). Numerous studies have demonstrated that teacher-delivered EI effectively produced positive outcomes for children with disabilities in inclusive settings (Coogle et al., 2018). To implement an EBP with fidelity and produce expected child outcomes, teachers need continuous professional development (PD) supports as they apply newly learned practices in the real world (Kraft et al., 2018).

Although increasing evidence supports the effects of PD on promoting teachers' use of EI (Snyder et al., 2018), limited research has examined such effects with inservice teachers in developing countries where early inclusive education is in its infancy. In China, compared to the percentage (i.e., 49.15%) of school-age students with disabilities having access to general education, only about 2% of under-school-age children with disabilities were enrolled in general education (National Bureau of Statistics of the People's Republic of China, 2007; The Ministry of Education of People's Republic of China, 2019). One of the most significant challenges for promoting early inclusive education in China is that teachers are not prepared and well-supported to teach children with disabilities in inclusive classrooms (Deng & Poon-McBrayer, 2019). The purpose of this study was to examine the effects of a PD intervention that combined training and a hybrid model of coaching on enhancing inservice preschool teachers' use of EI for children with disabilities in a polite early inclusion program in Beijing, China.

EI is a naturalistic and multicomponent approach for providing intentional and systematic teaching episodes to address children's individualized learning needs through classroom activities and routines (Snyder et al., 2013). Scholars emphasize that EI is an umbrella term for a group of practices related to instruction planning, implementing, and evaluating (Snyder et al., 2018). Essentially, children receive EI via teachers' presentations of embedded instruction learning trials (EILTs; Bishop et al., 2015). An EILT involves (1) a planned or naturally occurring antecedent that sets the occasion for (2) a child to demonstrate a target behavior, which is followed by (3) a planned or naturally occurring feedback or consequence (Rakap, 2017). This antecedent-behavior-consequence model (Cooper et al., 2020) describes the relationships between behavior, its environmental context, and its consequence. It is often used as a mechanism for behavior change by altering the events happening before and after the behavior. Prior research has found strong linkages between EI and improved outcomes for preschoolers with disabilities (Snyder et al., 2015). The documented positive children outcomes include preacademic (e.g., Botts et al., 2014), communication (e.g., Coogle et al., 2018, 2019), motor (e.g., Authors, 2000), adaptive (e.g., Toelken et al., 2012), and cognitive (e.g., Authors, 2000). However,

to our best knowledge, no peer-reviewed study has investigated the effects of EI on preschoolers with disabilities placed in inclusive classrooms in China.

Early childhood teachers have reported learning EI during their preservice training or inservice PD opportunities; however, teachers implemented these practices infrequently and with limited fidelity in inclusive classroom settings (Rahn et al., 2019). A growing literature has reported the effects of coaching-centered PD on improving teachers' learning and application of EI. For example, Rakap (2017) verified the causal relation between a training plus coaching intervention and preservice teachers' increased accurate use of EILTs in inclusive preschool classrooms. Snyder et al. (2018) described the effectiveness of a coaching-based PD intervention in promoting inservice preschool teachers' use of two EI practices: quality of goal writing and frequency of EILTs delivery. Coogle et al., (2018, 2019) validated the effectiveness of coaching-based PD in increasing teachers' implementation of embedded instruction practices. Despite the promising outcomes of using coaching-centered PD to support teachers' utilization of EI practices, a need exists to explore the variations of coaching (Snyder et al., 2018), including varied coaching agencies (e.g., expert coach, peer coach, self-coaching), coaching delivery methods (e.g., in-person, virtual, real-time, delayed), and the various dose of coaching. Furthermore, with the increasing interest in expanding the implementation of EI practices internationally (Embedded Instruction for Early Learning, 2020), more research is needed to examine the effects of coaching-centered PD on teachers' use of EI in inclusive classrooms of diverse sociocultural and linguistic contexts.

In the global movement of inclusive education, advocates for inclusion in China face similar challenges as their international counterparts. Chinese teachers and parents had shared concerns about the lack of individualized instructions or interventions for children with disabilities in inclusive classrooms (Deng & Poon-McBrayer, 2012). Researchers and early childhood education program administrators agree on the urgency to equip and support teachers with EBPs to promote successful inclusion (Malinen, 2013). Nevertheless, limited research has explored using EBPs to support children with disabilities in inclusive educational settings in China. No peer-reviewed literature has reported an empirical examination of how coaching-based PD can support inservice teachers' use of EI in Chinese inclusive preschool classrooms.

In this study, we evaluated the effects of a PD intervention on inservice preschool teachers' use of EILTs to address individualized objectives in preschoolers with disabilities in inclusive classrooms in Beijing, China. The PD intervention was centered on practice-based coaching (PBC). PBC is a well-recognized coaching framework in the field of early care and education (Early Childhood Learning & Knowledge Center [ECLKC], 2014; Snyder et al., 2015). The PBC framework operationalizes coaching as a three-component cyclical process, including establishing shared goals and action planning, conducting focused observations, and providing reflection and feedback. PBC also centers on collaborative partnerships between the coaches and coachees. Increasing number of studies have reported evidence that PBC is effective to increase teachers' effective use of EBPs—examples are embedded instruction practices (Snyder et al., 2018), positive behavior support related EBPs (Hemmeter et al., 2016), discrete trial training skills (Mason et al., 2017), individualized

behavior support plan (Chazin et al., 2018), and responsiveness strategies (Romano & Woods, 2018). The following research questions guided the investigation: (1) What effect does the PD intervention have on teachers' implementation fidelity of EILTs? (2) To what extent is the generality of teachers' use of EILTs? (3) Are teachers' use of EILTs associated with children's correct responses? (4) To what extent are the PD intervention and EI socially valid?

Method

We conducted a multiple-baseline single-case research design (Gast et al., 2018) across three teacher participants. We systematically manipulated the independent variable, included at least three sessions within each condition, measured each outcome variable across all phases and cases using more than one coders, and established interobserver agreement (IOA, Kratochwill et al., 2013). We included three replications across three teachers and met all requirements of What Works Clearinghouse (2020) Standards 4.1 with reservations.

Participants and Settings

After obtaining Institutional Review Board approval from the 2nd author's university, we recruited three teachers who were working in a private inclusive preschool in Beijing, China. The preschool director nominated the teachers based on the following criteria: (1) a lead teacher of an inclusive classroom, (2) has one study-eligible child in their classroom, (3) has not received prior training on EI, and (4) expresses interest and consent to receive PD to learn about and implement EI. We recruited three children with disabilities, one child from each teacher's classroom. The inclusion criteria for child participants comprised: (1) regular attendance (at least three days/week), (2) with a medical record of a diagnosis of disability issued by certified doctors, (3) with an individualized education program (IEP), (4) with no severe challenging behaviors, (5) receive no other services to address the target behaviors identified for this study, and (6) has parental consent to participate.

Three teacher–child dyads were formed: Anan-Andy, Bin-Bobo, and Chen-Cong. Anan was a 21-year-old female with an Associate Degree in early childhood education and had four-year experience working in inclusive education settings. Anan's classroom had three children with disabilities and 11 typically developing children, with a 14: 3 child: teacher ratio. Andy was a five-year-nine-month-old male diagnosed with autism spectrum disorder (ASD), and he participated in the afternoon in Anan's classroom. Bin was a 27-year-old female with an Associate Degree in special education and had two-year experience working in inclusive education settings. Bin's classroom had three children with disabilities and eight typically developing children, with a 11: 3 child: teacher ratio. Bobo was five-year-six-month-old male diagnosed with ASD, and he attended Bin's classroom full day. Chen was a 24-year-old female with an Bachelor's degree in special education and had three year experience working in inclusive education settings. Chen's classroom had three children

with disabilities and eight typically developing children, with a 11:3 child: teacher ratio. Cong was five-year-six-month-old male diagnosed with ASD, and he participated in the morning in Chen's classroom.

Professional Development Providers

The coaching team—two expert coaches (i.e., the 1st and 2nd authors) and two internal coaches—delivered the PD intervention. The first author was a doctoral student in early childhood special education (ECSE) with a minor in applied behavior science at a US university. The second author was an Assistant Professor and lead faculty of an ECSE program at a university in Beijing, China. Both expert coaches were familiar with coaching strategies and had experiences providing PD on EI related knowledge and skills to inservice teachers. The internal coaches were curriculum directors in the preschool. Yin had a bachelor's degree in special education and three-year experience teaching in inclusive preschool classrooms. Zan had an associate degree in early childhood education, five-year experience teaching preschoolers, and three-year experience teaching in inclusive preschool classrooms. Yin and Zan worked with teacher participants on a daily basis.

Settings

We conducted the study at a private preschool in Beijing, China. The classroom daily routine and activities included arrival, free play (FP), outdoor (ODR), breakfast, big group activity (BG), free play (FP), snack, BG, lunch, nap, snack, BG, FP, and departure. The setup of the classrooms was typical for Chinese preschools. Unlike regular preschool classroom arrangements in the western cultures, all major teacher-led big group activities occurred at a center area of a child-size table surrounded by child-size chairs. The only center was a bookshelf area for reading. Free play occurred mostly outside or in the bookshelf area. During the follow-up sessions, the program relocated to a brand-new building; however, the teacher–child ratio and the classrooms' setup remained similar.

Independent Variable

The independent variable (IV) was a PD intervention that combined one initial training session (InT; approximately 3 h) and three to six ongoing coaching sessions (OgC; approximately 30 to 60 min per session). The numbers of OgC received by each teacher are reported in *Results*. The coaching team delivered the InT session to each teacher participant individually via Zoom meeting. Two expert coaches joined the meeting remotely while two internal coaches and the teacher jointed the meeting together, sitting in a conference room of the preschool. One week before the InT, the teachers were provided with a *Teacher Manual* (i.e., written version of the training content with related resources; developed by the coaching team) and asked to view the manual. The InT involved an interactive PowerPoint presentation on EI and the

coaching process. The EI segment comprised of a) definitions and research base of the multiple component EI practices, b) video examples and non-examples of EILTs – the focused component of EI practice that the teachers would be working on during the intervention, c) teacher's completion of the *EILTs Planning and Implementation Forms* (See Supplementary Material A) with the coaching team's assistance, d) teacher's roleplay and reflection on EILTs process, e) coaches' feedback, and g) discussion. The coaching segment comprised of a) strategies for establishing and maintaining a collaborative partnership between coach and teacher and b) components and procedures of PBC. After learning about the knowledge and practices of EI and PBC, the teachers reviewed and signed the *Teacher-Coach Contract* (i.e., translated and adapted from the sample coaching contract [ECLKC, 2015]). Last, teachers completed the *Teacher Needs Assessment and Action Plan Form* (See Supplementary Material B) to identify an initial goal related to the use of EILTs to promote children's target behaviors. During the OgC sessions, the coaches focused their observations and debriefings on supporting teachers to achieve those initial goals.

Each ongoing coaching (OgC) session included an observation (at least 15 min) and a debriefing (at least 15 min). Internal coaches and expert coaches provided OgC in alternating order. Internal coaches provided in-person debriefings immediately following onsite observations. Expert coaches observed teacher–child interactions via video recordings and provided teacher a delayed debriefing within three days after the date of video recording. Coaches only provided coaching during the planned OgC sessions and used the *Coach Observation and Debriefing Form* (See Supplementary Material C) to guide the process. The debriefing sessions included: opening the meeting, teacher self-reflection, supportive feedback, corrective feedback, future goals and action plans, and closing the meeting (Rakap, 2017). Coaches used the *EILTs Implementation Fidelity Coding Form* (See Supplementary Material D) to assess teacher's implementation fidelity of EILTs and facilitate the debriefing process.

Strategies to Promote Generality of EILTs

In this study, we planned and implemented several strategies for promoting generalizations of teachers' use of EILTs based on Cooper et al.'s (2020) suggestions. First, we modeled different ways to deliver EILTs during various classroom routines and activities as a means to teach enough exemplars. Second, we taught loosely by introducing a wide range of both antecedent and consequence-based strategies from which teachers could choose and implement EILTs based on real-time circumstances. Third, we used video clips of teachers' actual interactions with focus children to teach how to deliver EILTs as a means to introduce common stimuli that reflect the real environments. Fourth, we provided expert and internal coaches to maximize teachers' contact with reinforcement.

Procedure Fidelity and Interobserver Agreement

The procedure fidelity of the PD intervention was self-assessed by the coaches for all participants and across all conditions. During baselines and follow-up phases, the coach team provided no training or coaching to the teachers. Specifically, the expert coaches had no contacts with participating teachers; two internal coaches observed their own and each other's provision of professional development to teacher participants and reported teachers received no training or coaching on EILTs. The procedure fidelity was 100% for all participants ($M = 100\%$, $SD = 0$) across baseline and follow-up phases. During intervention phases, the coach team used the *Initial Training Session Fidelity Checklist* (See Supplementary Material E) and the *Ongoing Coaching Session Fidelity Checklist* (See Supplementary Material C) to self-report procedure fidelity (i.e., $Mean = 100\%$, $SD = 0$ for all teachers' InT and OgC sessions). The independent reliability coders viewed 100% video recordings of InT sessions and at least 33% (i.e., 33% to 40%) of the OgC sessions to evaluate the procedure fidelity of the PD intervention. The mean IOA of procedures fidelity was 100% (SD = 0), 100% ($SD = 0$) across all participants for the InT session and 100% ($SD = 0$) for the OgC sessions, across all participants.

Dependent Variable

The primary dependent variable (DV) was the accuracy of EILT implementation and was measured by the percentage of EILTs correctly implemented by the teachers during the activities and routines selected by coaches and teachers together based on children's attending hours and the characteristics of target behaviors. The selected activities and routines to embed EILTs for both Anan-Andy and Bin-Bobo were transitions, big group activities, and dinner time; for Chen-Cong, it was transitions, outdoors, big group activity. To correctly implement an EILT and score "1" (See example of teacher's correct implementation of an EILT in Table 1), teachers needed to deliver both antecedent-based strategies (ABSs) and consequence-based strategies (CBSs) correctly. During the InT, the teachers learned about four types of ABSs including (1) environment arrangement (Ledford et al., 2018), (2) provide choice (Wong et al., 2014), (3) provide highly preferred items or activities (Wong et al., 2014), and (4) verbal instruction (Ledford, et al., 2018). Teachers and coaches discussed which ABSs were most likely to evoke target behaviors for the participating children. The onset of an EILT was determined when the teacher was observed implementing one or more of those four ABSs intended to evoke children's target behaviors. Based on the child's response following the ABS, the teacher responded to the child using CBSs. During the InT, teachers learned about how to provide specific CBSs according to different levels of children's responses. Prompts involved verbal, gestural, or physical assistance given to the learner to help them acquire or engage in a targeted behavior or skill (Wong et al., 2014). Table 1 presents the definitions of children's target behaviors and examples of EILTs. Children's behaviors were coded and graphed to examining the associations between teachers' implementation of EILTs and children's outcomes.

Table 1 Children Target Behaviors and Examples of EILTs

Target behavior	Antecedent-based strategies	Child response	Consequence-based strategies
Andy Child follows group instruction within 5 s of the instruction	The storytime is done, teacher says to the class, "Time to wash hands. Line up, please."	【+】Andy lines up within 5 s of the instruction	Teacher says, "Awesome job listening and lining up!" and gives Andy a high five
		【–】Andy still sits on his chair after 5 s of the instruction	Teacher points to the line while looking at Andy and walking towards him
Bobo Child verbally requests an item or activity from an adult using a complete sentence	At lunchtime, teacher holds on to the Bobo's spoon	【+】Bobo asks, "I want my spoon."	Teacher says, "Nice job asking for your spoon!" and gives Bobo his spoon
		【–】Bobo sits on his chair and stares at the food	Teacher looks at Bobo and says, "If you need anything, you can ask for it."
Cong Child articulately responds to teacher-initiated questions more than once, with appropriate eye contact	Teacher is showing a puppet dog to the class and asks Cong a series of questions: "What's the color of the dog?" "How many ears does it have?" "Do you have a dog?"	【+】Cong correctly responds to the questions more than once, all with clarity and appropriate eye contact	Teacher says, "You spoke so clearly, and you looked at me! Wonderful!" and gives Cong a sticker
		【–】Cong mumbles to the question and looks at the window	Teacher directs Cong's chin, looks into his eyes, and repeats the question

+ = correct response; – = response needs improvement

During the follow-up sessions, the original three child participants had graduated, and the preschool program relocated to a new building. The generality of teachers' use of EILTs were examined via three aspects: (1) response maintenance when the intervention was removed, (2) with novel children with disabilities, and (3) within novel settings. During the follow-up sessions, the teachers nominated one to two new children with disabilities in their classrooms and identified their individualized learning objectives. New coding rules of correct implementation of EILTs were established based on novel child target behaviors.

Teacher Implementation Fidelity, Child Outcomes, and IOA

Data collectors videotaped the interactions between participating teacher–child dyads during scheduled data collection days. After a video recording was completed, a primary coder (i.e., the first or second author) coded the video to assess teachers' implementation of EILTs. The occurrences and accuracy of EILTs were coded based on manualized coding rules for each teacher–child dyad. A reliability coder who was not involved in field data collection and blind to the study conditions coded the same video independently and concurrently (i.e., within a week of the primary coder completing the coding) using the same set of rules for each dyad. IOAs, calculated by the trial-by-trial method (Cooper et al., 2020), were established for the percent of correct EILTs and the percent of correct child responses for at least 33% of the sessions across all conditions for all teacher–child dyads. For Anan, the IOA mean was 89.64% (ranged from 86.96% to 92.31%) during baseline, 86.57% (ranged from 78.95% to 100%) during intervention, and 100% during follow-up. For Bin, the IOA mean was 100% (ranged from 100 to 100%) during baseline, 83.55% (ranged from 78.95% to 84.65%) during intervention, and 87.50% during follow-up. For Chen, the IOA mean was 88.54% (ranged from 83.33% to 100%) during baseline, 78.41% (ranged from 75 to 81%) during intervention, and 100% during follow-up. For Andy, the IOA mean was 83.61% (ranged from 82.61% to 84.62%) during baseline and 84.30% (ranged from 78.95% to 90.91%) during intervention. For Bobo, the IOA mean was 92.59% (ranged from 77.78% to 100%) during baseline and 81.95% (ranged from 76.92% to 84.21%) during intervention. For child Cong, the IOA mean was 89.58% (ranged from 75 to 100%) during baseline and 78.41% (ranged from 75% to 81.81%) during intervention.

Procedures

Internal Coach Preparation

Internal coach preparation began two months before the onset of the baseline. The internal coaching preparation comprised six 60 to 80 min online training sessions via Zoom, with homework assigned between sessions (See Supplementary Material F). The expert coaches provided interactive PowerPoint presentations to introduce key features of EI practices and the PBC process to the internal coaches. For

active ingredients of an EILT (i.e., ABSs, CBSs) and PBC (e.g., collaborative partnership strategies, debriefing tips, and usage of observation and debriefing forms), expert coaches provided both video and in-person modeling. The expert coaches then engaged in roleplay with the internal coaches following their reflections and expert coaches' feedback. After each session, internal coaches reviewed what they learned and previewed the next session. Both internal coaches reached 100% fidelity for delivering OgC during roleplays before the intervention. During the intervention, the expert coaches continued to provide one-on-one assistance to the internal coaches to address questions and maintain their implementation fidelity. Expert coaches' assistance involved (1) reviewing internal coaches' debriefing sessions and coaching fidelity checklists and (2) meeting with internal coaches to reflect on previous OgC sessions. The coaching team also communicated regularly via WeChat, a messenger application widely used in China.

Child Target Behavior Selection

The coach team, teacher, and parents reviewed all learning objectives listed on the target child's individualized educational program (IEP) together and made shared decision on which learning objective to be prioritized and targeted for the current intervention. All three children received independent assessments from external early intervention service providers using the *Verbal Behavior Milestones Assessment and Placement Program: a language and social skills assessment program for children with autism or other developmental disabilities* (VB-MAPP; Sundberg, 2008). The learning objectives on a child's IEP were primarily drawn from VB-MAPP results. The coach team, teacher, and parents co-selected a child's target behavior based on five criteria: (1) VB-MAPP results suggested the need to improve the behavior and thus the behavior was included in the child's IEP, (2) teacher and parent agreed on the priority of the target behavior, (3) teacher reported that the child was not able to perform the target behavior independently and reliably at the onset of the study, (4) the behavior was not addressed by external intervention or educational services, and (5) the behavior is appropriate for embedding into ongoing activities and routines of the child's classroom.

Baseline

In baseline, teachers were instructed to teach target behaviors to child participants, but no further information was provided. Particularly, the coach team provided no training or coaching to teachers regarding using EILTs to address child target behaviors. The videotapes were viewed and coded to determine teachers' implementation of EILTs and children's responses. The researchers determined when to enter intervention for each teacher based on the stability of teachers' behaviors (i.e., percent of correct EILTs) during baseline.

Intervention

The first teacher who demonstrated a stable baseline performance—no upward trend of the last three data points and no greater than 20% difference per the level of data—was selected to enter the PD intervention by receiving the *Teacher Manual* and attending the InT session. The same intervention procedures were repeated with the next two teachers when their baseline data demonstrated stability. The internal and expert coaches provided the OgC session in alternating order.

Follow-up

The follow-up sessions occurred three months after the intervention's closure to examine three aspects of the generality of teachers' use of EILTs: response maintenance, setting generalization, and response generalization. In the present study, we examined the generality of each teacher's use of EILTs with novel children with disabilities (i.e., response generalization), in a novel classroom (i.e., setting generalization), when the intervention was removed (i.e., response maintenance). The teachers participated in no training and OgC sessions since the termination of the intervention.

Data Collection and Analysis

Data collectors were 12 undergraduate students taking a Behavior Observation class with the second author. They were trained with standard procedures (i.e., task analysis for video recording and video storage processes) to videotape interactions between teacher–child dyads during pre-selected activities and routines across all experimental conditions. The average length of videos recorded across baseline and intervention was 54 min for Anan, 48 min for Bin, and 44 min for Chen. All videotapes were viewed and coded by a primary coder using the *EILTs Implementation Fidelity Coding Form* based on sets of coding rules of each teacher–child dyad. Within a week of the primary coding, a trained reliability coder who was blind to experimental condition coded the same recording independently to establish IOA. Two of the second author's graduate research assistants received IOA training. During the first week of baseline phases, the reliability coders had to demonstrate more than 85% IOA for three 20-min training clips of each participating teacher's interactions with target child before coding independently. The primary coders and the reliability coders attended a weekly calibration meeting to maintain reliability during the period in which they were actively coding videos. More than 33% of videos across all conditions for each teacher were coded by reliability coders. Primary data were graphed for visual analysis immediately after coding to determine phase changes for each teacher–child dyad and evaluate the PD intervention effects. Additionally, Tau-U (Parker et al., 2011), a nonparametric effect size calculation method was used at the end of the study to determine the effect of the PD intervention on

teachers' use of EILTs. The Tau-U calculations were conducted on a web-based application (Vannest et al., 2016).

Social Validity

Social validity was assessed using a revised and translated version of *Intervention Rating Profiles* (IRP; Rakap, 2017). The *Intervention Rating Profile-Revised-PD* (IRP-R_{PD}) assessed the extent to which the PD intervention procedures were acceptable, feasible, and satisfactory to internal coaches and teachers. The *Intervention Rating Profile-Revised-EI* (IRP-R_{EI}) assessed the level of acceptability, feasibility, and effectiveness of EI in addressing children's individualized learning objectives in inclusive classrooms. Both measures were a 10—item, 6—point Likert scale (i.e., 1 = strongly disagree and 6 = strongly agree). Internal coaches and teachers completed both measures anonymously upon the termination of the intervention.

Results

All inservice teachers' implementation fidelity of EILTs increased after they entered the PD intervention. Figure 1 (left panel) depicts the percentages of teachers' correct implementation of EILTs across baseline, intervention, and maintenance/generalization conditions. During baseline, the percentages of teachers' correct implementations of EILTs were low, and they increased immediately after the InTs were provided for all teachers. Changes in level across conditions exceeded the changes in level within conditions. Minimum overlapping data were observed between baseline and intervention conditions across all participants. Additionally, the effect size statistics (i.e., Tau—U statistics) for each participant and across all participants for each condition comparison were significant. The Baseline vs. Intervention Tau-U comparison statistics are 0.84 ($p < 0.05$) of Anan, 0.86 ($p < 0.01$) of Bin, 1.00 ($p < 0.01$) of Chen, with an omnibus score of 0.90 ($p < 0.001$). The Baseline vs. Maintenance/Generalization Tau-U comparison scores are 1.00 ($p < 0.01$) for all three teachers with an omnibus score of 1.00 ($p < 0.001$). There were three repetitions of the experimental effect across three participants at three different points in time. Thus, these data demonstrate a functional relation between the PD intervention and teachers' implementation fidelity of EILTs and meet the single case research design standards recommended by Kratochwill et al. (2013). Figure 1 (right panel) depicts the percentages of children's correct responses across experimental conditions. Associations between teachers' implementation fidelity and children's improved performance (i.e., non-experimental corollary outcomes) were also observed.

Dyad 1: Anan and Andy

As shown in Fig. 1 (left panel), during the baseline, we observed an average of 2% correct implementation of EILTs. Anan's implementation was somewhat variable

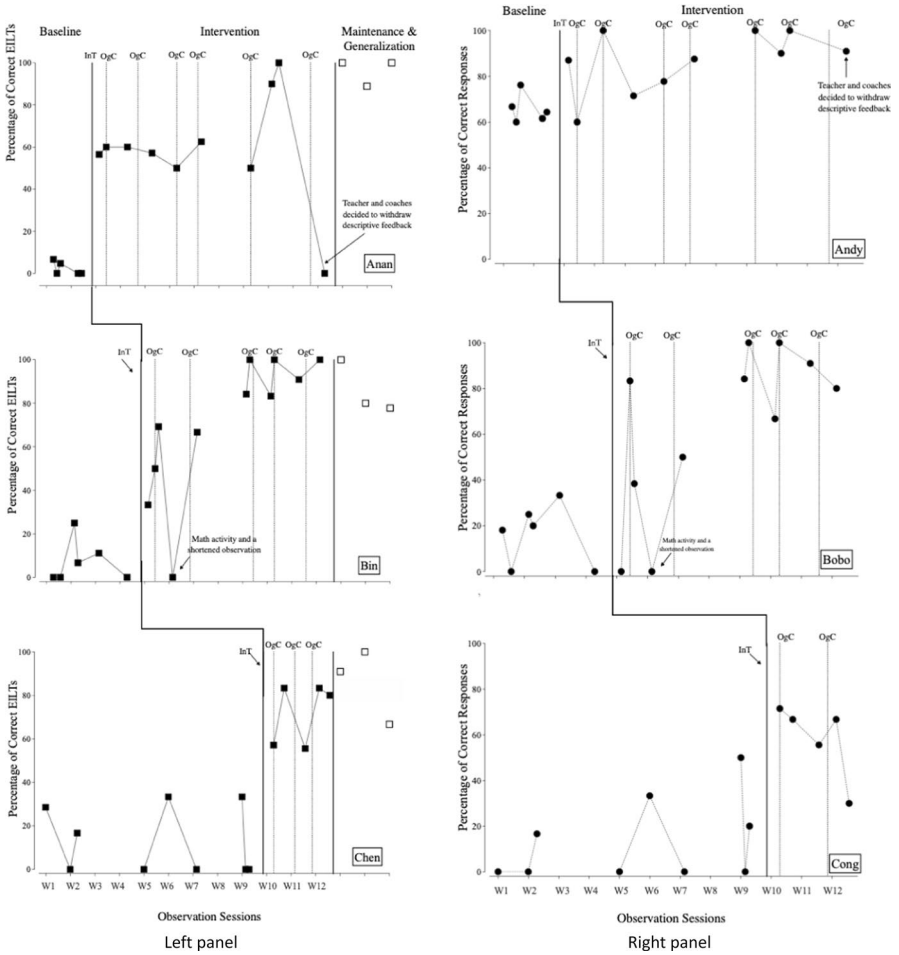


Fig. 1 Inservice Teachers' Use of Embedded Individualized Learning Trials (*Left Panel*) and Children's Individualized Target Behaviors (*Right Panel*). The program was closed for 11 consecutive days due to unexpected reasons; the planned data collection did not occur for all dyads. For Chen-Cong, the scheduled data collection between did not occur for 19 consecutive days due to Cong's unexpected sick leave. InT = initial training session; OgC = ongoing coaching session; W1 = week 1

(*SD* = 0.03%) with a decreasing trend. After the InT session, an immediate change in the level of correct implementation (i.e., 0% to 57%) of EILTs was observed, and an increasing trend with some variability (*M* = 59%; *SD* = 26%) was observed during the intervention condition. During the InT, Anan identified CBS as her focused goal, which involved providing Andy with appropriate prompts and reinforcement based on his response. During each observation and debriefing session, the coaching team prioritized supporting Anan's practice of CBS. As a result, as Anan improved on delivering proper CBS, we observed an increasing level of correct EILT implementation. Anan received six OgC sessions. During the last OgC session, Anan reflected

that Andy had mastered the original learning objective by following group instructions consistently and reliably, and she would want to move on to Andy's next learning objective. The coaching team reviewed Andy's data and affirmed Anan's advocacy. The coaching team suggested Anan withdraw descriptive feedback on Andy's correct response. Thus, based on the Anan-Andy dyad's coding rules, Anan demonstrated no correct use of EILT during the observation session that occurred after the last OgC session. During the maintenance/generalization condition, Anan's use of EILTs ($M = 96\%$; $SD = 6\%$) with novel children with disabilities and within the novel setting were at levels comparable to the intervention condition with no data points overlapping with baseline sessions. Andy demonstrated a relatively high level of correct response ($M = 66\%$; $SD = 6\%$) during the baseline. After Anan received PD intervention, Andy's mean correct responses to group instructions increased to 86.46% ($SD = 13\%$). Particularly, Andy's correct response rate remained stable even when Anan withdrew descriptive feedback on his correct responses.

Dyad 2: Bin and Bobo

During the baseline of Bin's use of EILTs, we observed an average of 9% correct implementation of EILTs, and her implementation was somewhat variable ($SD = 10\%$). With the introduction of the InT session, an immediate change in Bin's correct implementation (i.e., 0% to 33%) of EILTs was observed, and an increasing trend with some variability ($M = 71\%$; $SD = 39\%$) was observed during the intervention condition. During the intervention, one scheduled observation was disrupted and thus shortened due to a preschool event. During the fourth intervention data collection session, Bin was leading a math-themed big group activity which explained why she delivered zero EILT to Bobo. In the InT session, Bin first identified using appropriate ABS as her initial focused goal, which mostly involved arranging the environment to create opportunities for Bobo to request and giving verbal instructions that prompt Bobo to request. After demonstrating increasing correct use of ABS, Bin advocated moving on to a new goal—working on CBS. Consequently, the coaching team focused their observation and debriefing sessions on CBS as requested by Bin. As Bin improved on delivering correct ABS and then CBS, we observed an increasing level of correct EILT implementation. Bin received five OgC sessions. During the maintenance/generalization condition, Bin's use of EILTs with novel children with disabilities and within the novel setting were at levels comparable to the intervention condition with no data points overlapping with baseline sessions. Bobo demonstrated a relatively low level of correct response ($M = 9\%$; $SD = 15\%$) during the baseline condition. After Bin entered the intervention, Bobo's mean correct responses to group instructions increased to 63.46% ($SD = 37\%$).

Dyad 3: Chen and Cong

During the baseline, Chen demonstrated an average of 12% correct implementation of EILTs, and her implementation was somewhat variable ($SD = 16\%$). After attending the InT session, an immediate increase in the level of Chen's correct

implementation (i.e., 0% to 57%) of EILTs was observed, and an increasing trend with some variability ($M = 72\%$; $SD = 14\%$) was observed during the intervention condition. During the InT, Chen suggested setting "effectively use CBS" as her focused goal, which primarily involved using the gestural prompt to guide Cong's eye contact and providing appropriate reinforcement based on his response. The coaching team focused their observation and debriefing sessions on Chen's practice of correct CBS. As a result, when Chen improved on delivering correct CBS, we observed an increasing level of correct EILT implementation. Chen received three OgC sessions. During the maintenance/generalization condition, Chen's use of EILTs ($M = 86\%$; $SD = 17\%$) with novel children with disabilities and within the novel setting were at levels comparable to the intervention condition with no data points overlapping with baseline sessions. Cong demonstrated a relatively low level of correct response ($M = 13\%$; $SD = 18\%$) during the baseline condition. After Chen entered the intervention, Cong's mean correct responses to group instructions increased to 58% ($SD = 17\%$).

Social Validity

Teachers and internal coaches indicated acceptance of both PBC and EI. Internal coaches showed slightly more favorable attitudes towards both PBC and EI compared with inservice teachers. For internal coaches, the mean rating across the ten items was 5.75 (range = 2—6) for IRP-R_{PBC} and 4.2 (range = 1—6) for IRP-$R_{EI.}$ The mean rating for inservice teachers was 4.84 (range = 2–6) for IRP-R_{PBC} and 3.47 (range = 1–5). Supplemental Material G presents scores for each IRP item for both PBC and EI from peer-coaches and inservice teachers.

Discussion

To address the first research question, this multiple baseline across participants single-case study revealed the effect of the PD intervention in increasing teachers' implementation fidelity of EILTs. The findings added cross-sociocultural evidence to the literature base (e.g., Artman-Meeker et al., 2015; Kraft et al., 2018; Kretlow & Bartholomew, 2010) that reports the effectiveness of PD on producing desirable teacher performance. The present PD intervention combined virtual training and ongoing coaching. Manualized PD materials in the Chinese language are ready-to-use for future researchers and teacher educators. The effectiveness of this intervention affirms the theories of implementation science in which training and coaching as two competency drivers (Fixsen et al., 2015) are essential mechanisms to develop, improve, and sustain teachers' ability to implement an intervention as intended to benefit children.

In the current study, the coaching team served as a unified entity to deliver the PD support. Particularly, expert coaches and internal coaches had shared responsibilities, used the same coaching procedures, and thus functioned as one intervention agent. To ensure the coaching team's implementation integrity, we prepared the

internal coaches before the intervention until they met the coaching fidelity standards. Throughout the intervention, we also monitored the coaching fidelity to ensure the designed coaching components were delivered with high integrity regardless of whether it was provided by the expert coach or the internal coach, virtually or in-person, instantly or delayed. The functional relation between the PD intervention and teachers' increased accuracy of EILTs provides evidence of the effectiveness of this hybrid coaching method. Such effectiveness is not surprising because the coaching topologies used in this model are variations that have been described theoretically and tested empirically in the teacher coaching literature. For example, Kraft et al. (2018) conducted a meta-analysis of the effect of PreK-12 teacher coaching and found 60 group design studies have reported causal evidence. Among those studies, 78% delivered coaching via an in-person mode and 22% via a virtual mode.

At its core, the present PD intervention is based on the PBC framework, which is the hallmark PD mechanism sponsored by the federal agency (ECLKC, 2014) to support the Head Start workforce in the USA. An increasing number of studies have reported the promising causal relation between PBC-based PD and improved teacher performance (Shannon et al., 2020; Snyder et al., 2018). However, few empirical studies conducted in contexts outside of the USA have reported the effectiveness of the PBC framework in supporting inservice preschool teachers' practices. This study is the first to use PBC-based PD for Chinese inservice preschool teachers working in inclusive classrooms. The findings provide promising evidence that can inform researchers and teacher educators preparing personnel for inclusive education in under-represented countries.

The results yielded some interesting findings regarding the varied progress trajectories of teachers' EILTs implementation fidelity. Although teachers' progress rate could be explained by the levels of difficulty in addressing different child target behaviors, we argue that further consideration is needed when solely using the researcher's definition of implementation fidelity to evaluate teachers' effective EI practices. Specifically, based on the Anan-Andy dyad's coding system, Anan had to provide descriptive feedback (e.g., "Good job following the instruction so quickly!") to Andy's correct response to score a correct EILT. During the intervention, Anan only provided descriptive feedback to Andy's correct responses for about half of the EILTs. However, we still observed Andy's steady improvement of correct responses since Anan entered intervention and especially when Anan withdrew descriptive feedback entirely during the last data collection point. This finding raises several questions. First, does researcher-defined implementation fidelity of EILTs underestimate teachers' effective use of EILTs? Second, to what extent is descriptive feedback necessary to produce children's behavior change in natural teaching contexts? Is general feedback (e.g., "Good job!") or other natural consequences of following group instructions sufficient to reinforce Andy's consistent exhibition of correct responses? More research is needed to examine the optimal definition of implementation fidelity of EILTs and the most significant component of EILTs. Additionally, future researchers may also conduct experimental studies to further disentangle the effectiveness of specific components of the PD (i.e., initial training plus ongoing coaching) package presented in this study. Such work is essential to bridge the research-practice gap by ensuring that researchers' definitions of implementation

fidelity of an EBP best reflect teachers' effective teaching in the real world (Cook & Odom, 2013).

Finally, we focused on EILTs—one key element of the multicomponent EI. More research is needed to examine teachers' use of other components of EI practices. For example, we found that specific activity or routine might not be the optimal time to insert EILTs for addressing particular child behavior. Bin was observed delivering zero EILTs during a data collection day when the primary activity was math themed. She was observed focusing on group instruction and only provided several individual teaching episodes for Bobo to address math-related skills. Another indication of teachers' improved understanding and practices of EI is to capture more appropriate learning opportunities (i.e., when to teach; Authors, 2016). We coded EILTs frequency across conditions for all teachers, but we did not observe an increased frequency of teachers' implementation of EILT. We excluded this data in the report because such observation was only complementary and not designed to address our core research questions. We argue that even though we did not observe the increased frequency of EILTs across conditions, teachers' use of EILTs might still have improved by identifying and utilizing more naturally occurred opportunities to embed instruction compared to that in baseline. However, the current data collection system was not designed to capture missing opportunities.

The majority of prior research (e.g., Coogle et al., 2018; Rakap, 2017) reported only one or two aspects of generality regarding teacher's use of EI practices (e.g., generalization in new settings, maintenance overtime). The generalized use of EILTs with new target child behaviors and new children with disabilities (i.e., response generalization) was least examined. To answer the second research question regarding generality of teachers' use of EILTs, we examined all three aspects of the generality of teachers' use of EILTs and the findings revealed positive outcomes for all three participating teachers. The results affirmed Cooper et al. 's (2020) strategies for promoting maintenance and generalization of behavior change.

However, it was unclear to what extent the internal coaches supported teachers' generalized EILTs with new children with disabilities. Recall we intentionally placed the internal coaches as non-contingent reinforcement for teachers to improve the generality of EILTs. From self-report, the internal coaches' supports to teachers between the intervention closure and the follow-up sessions did not resemble the PD intervention in terms of dosage of observation and debriefing. Future research should add an independent assessment of internal coaches' sustained use of the PD strategies to obtain an accurate picture of the sustainable use of the PD intervention and how it affects the sustainability of teachers' use of EBPs. Overall, our experimental design for evaluating the maintenance and generalization of teachers' use of EILTs met WWC's standards with reservation because at least three data points were collected during the follow-up sessions. However, to further strengthen the causal relation between the PD intervention and teachers' generality of EILTs, more observations are needed across a more extended period of time.

Regarding the third research question on children's behavior changes, we observed positive children outcomes associated with increased teachers' implementation fidelity of EILTs across all three child participants. As we summarized in the introduction section, numerous empirical studies have proved the effects of

EI practices on producing positive outcomes of children with disabilities in inclusive settings. This study is the first attempt to examine the impact of the inservice teacher delivered EI on improving quality early inclusion of Chinese preschoolers with disabilities. It is important to acknowledge that the participating children were considered having mild to moderate disabilities. More research is needed to include children with more severe support needs. Additionally, the positive association between teachers' performance and children's outcomes was not causal, given that teachers' delivery of EILTs was not a consistent independent variable. Future research may examine such causal relations after teachers have reached the criteria of EILT implementation fidelity.

Noticeably, Andy exhibited a relatively high percentage of correct behaviors during the baseline. Recall the procedure for selecting a child's target behavior; we relied mainly on Andy's VB-MAPP results and Anan's report. One possible explanation for Andy's high level of compliance in the baseline is that Andy's target behavior's operational definition allowed too much time for Andy to respond. However, Anan's expectation for correct behavior might be Andy responds quicker. We chose to maintain the definition of Andy's behavior for two reasons. First, Andy still exhibited the need for additional support on "following group instructions" compared to typically developing children. Second, the coding system for Anan-Andy has already been established. The coders have been trained and reached reliability criteria, maintaining the original operational definition was a decision driven by cost-effectiveness.

Finally, we discuss to what extent the PD intervention and EI socially valid. A noticeable finding of this study is the discrepancy between internal coach and inservice teachers' perceptions of PD intervention. Overall, internal coaches showed more favorable attitudes toward both PBC and EI than inservice teachers. One explanation is that as members of the coaching team, internal coaches received more training and invested more time on PBC and EI; thus, they were more experienced and had more ownership of the project. Inservice teachers indicated stronger disagreement on "EI only requires little training and technical skills." The amount of training received on EI may also result in the discrepancy between internal coach and teacher participants' perceptions of the technical skills required to implement EI with fidelity. To increase inservice teacher buy-in (Fixsen et al., 2015), future researchers or PD developers should consider strategies to increase teachers' sense of ownership, tailor the technical content of EBPs to better fit into teachers' repertoire and the contexts on the job. Finally, given the importance of sustainable usage of the EI after the PD intervention was withdrawn, researchers may gather information from the participants regarding which specific components of the PD package and the EI they may most likely continue to use in their daily practice.

Limitations

We acknowledge several limitations in this study. First, only three teacher–child dyads were included in this single-case design study. Systematic replications (e.g., extend the intervention to different settings or children with other individual needs) or group design studies at a larger scale should be used to ascertain the generalizability of these results as well as the efficacy of the PD intervention. Second, other complementary child assessments are needed to better understand the linkages between the PD intervention and teachers' use of EI practices on children's outcomes.

Conclusion

Within the flourishing area of research demonstrating the efficacy of coaching-based PD interventions carried out by trained teachers in educational contexts in increasing students' outcomes, this study makes an original contribution to the existing literature by focusing on the effects of this kind of intervention on inservice preschool teachers and children with disabilities in underrepresented sociocultural contexts. Overall, this study provides additional evidence of using training and coaching to improve teachers' implementation fidelity of embedded instruction practices and promote meaningful inclusion for young children with disabilities.

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Author Contributions JA designed the study, collected and analyzed the data, and wrote the manuscript. MZ collected and analyzed the data. SB critically reviewed the manuscript. EH critically reviewed the manuscript and advised on the design and implementation of the study.

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Declarations

Conflict of interest The authors declare that there are no competing interests.

Ethical Approval and Consent This study was approved by an equivalent office to the Institutional Review Board in the second author's university. All teacher participants and the parents of child participants provided written consent for participating in this study.

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